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Replaceable Cutting Head Having Mounting Portion with Driven Projections, Tool Holder and Rotary Cutting Tool

Abstract

A cutting tool includes a cutting head including a rearward mounting portion and a head through hole. The mounting portion includes a male coupling member having a non-cylindrical, conically shaped head abutment surface, and at least two driven projections. The cutting tool includes a tool holder having a holder pocket which includes a female coupling member having a non-cylindrical, conically shaped forward holder abutment surface, at least two driving recesses and an internal threaded hole. In a locked position, the male coupling member is removably retained in the female coupling member by a fastening screw located in the head through hole and threadingly engaged with the holder internal threaded hole. The head abutment surface abuts the forward holder abutment surface thereby forming a secure coupling between the male and female coupling members. The driven projections are located in the driving recesses for providing torque transfer between the two parts.

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Background/Summary

RELATED APPLICATIONS [0001] The present application claims priority to U.S. Provisional Patent Application No. 63/558,764 filed 2024 Feb. 28. The contents of the aforementioned application are incorporated by reference in their entirety.

FIELD OF THE INVENTION

[0002] The subject matter of the present application relates to rotary cutting tools of the type in which a replaceable cutting head, having a male coupling member, is removably retained in a female coupling member of a tool holder, by means of a coupling mechanism having a taper. More particularly it pertains to a coupling mechanism which also has at least two inter-engaged projections and recesses, for torque transfer, and yet further in particular where the coupling mechanism has an additional fastening screw.

BACKGROUND OF THE INVENTION

[0003] Rotary cutting tools can be provided with a coupling mechanism for releasably retaining a replaceable cutting head within a tool holder by a fastening screw. An example of such a rotary cutting tool is disclosed in, for example, U.S. Pat. No. 7,431,543, showing a machine reamer which includes a base and an interchangeable reamer head assembled on the base. The reamer head is clamped with a conical shoulder into a complementary face-side insert seat of the base by means of a tension rod to provide the necessary coaxiality. The tension rod has a complementary polygonal section extending through an axis-central polygonal opening of the indexable head, so that a good rotational catch is guaranteed.

[0004] Another example is U.S. Pat. No. 7,004,692 showing a cutting head, a screw member and a tool shank. The screw member is in threaded engagement with the tool shank. The cutting head is centered with respect to the tool shank with a conical portion of the cutting head located in a conical forward portion of the tool shank. The cutting head has a head bore having a plurality of locking wings and the screw member has a plurality of clamping wings. Rotational coupling between the cutting head and the screw member is obtained by the engagement of the clamping wings of the screw member and stoppers which protrude forwardly in the axial direction from the locking surfaces of the locking wings. That is to say, torque is transferred from the tool shank to the head via the clamping wings on the screw member.

[0005] Yet another example is U.S. Pat. No. 9,919,366 showing a cutting insert having a first coupling structure, a tool holder having a complementary second coupling structure with a seat for a milling tool system, and a milling tool system. The first coupling structure has a circular cylindrical element with a cylinder abutment surface, the circular cylindrical element projecting axially from a bottom surface of a central body and being arranged concentrically with respect to a central axis. A perpendicular abutment surface is arranged perpendicular to the central axis, and at least two ribs project axially from the bottom surface. The ribs are symmetrical with two convexly curved flank surfaces which serve as driven surfaces. Each rib has a contact area with a normal direction at least essentially perpendicular to the central axis for opposing a torque about the central axis.

[0006] It is an object of the subject matter of the present application to provide a coupling mechanism of a replaceable cutting head in a tool holder which is suitable, in particular, for rotary

cutting tools having a small outer cutting diameter.

SUMMARY OF THE INVENTION

[0007] In accordance with a first aspect of the subject matter of the present application there is provided a replaceable cutting head, for rotary cutting operations, having a head central axis, defining opposite forward and rearward directions, and opposite rotationally forward and rearward directions with the rotationally forward direction being the cutting direction, the replaceable cutting head comprising: [0008] a forward portion forming a cutting portion and a rearward portion forming a mounting portion; and [0009] opposite head forward and rearward surfaces and a head peripheral surface extending therebetween, the head forward surface being located at the cutting portion and the head rear surface being located at the mounting portion; [0010] a head through hole comprising a through hole peripheral surface which extends about a through hole central axis, and intersects the head forward and rearward surfaces; wherein: [0011] the mounting portion comprises: [0012] a male coupling member protruding rearwardly from a head base surface, the head base surface extending transversely with respect to the head central axis, and defining a boundary between the cutting portion and the mounting portion; and [0013] at least two angularly spaced apart driven projections projecting from the head rearward surface, each driven projection comprising a projecting wall comprising two opposite and angularly spaced apart projection flank surfaces, including a rotationally leading and a rotationally trailing projection flank surface, the rotationally trailing projection flank surface comprising a driven surface, for opposing a torque about the head central axis; wherein: [0014] the head base surface comprises a head axial abutment surface; and [0015] the male coupling member comprises a non-cylindrical, conically shaped head abutment surface.

[0016] In accordance with a further aspect of the subject matter of the present application, there is also provided a tool holder, having a holder longitudinal axis defining opposite forward and rearward directions, and opposite rotationally forward and rearward directions with the rotationally forward direction being the cutting direction, the tool holder comprising a holder pocket extending rearwardly from a holder forward surface, the holder forward surface extending transversely with respect to the holder longitudinal axis; wherein: [0017] the holder pocket comprises: [0018] a female coupling member recessed in the holder forward surface, female coupling member comprising: [0019] a non-cylindrical, conically shaped holder abutment surface; and [0020] a female member bottom surface; [0021] at least two angularly spaced apart driving recesses recessed in the female member bottom surface, each driving recess comprising a recess circumferential wall comprising two opposite and angularly spaced apart recess flank surfaces including a rotationally leading and a rotationally trailing recess flank surface, the rotationally trailing recess flank surface comprising a driving surface, for providing a torque about the holder longitudinal axis; and [0022] a holder internal threaded hole recessed in the female member bottom surface; wherein: [0023] the holder forward surface comprises a holder axial abutment surface.

[0024] In accordance with a yet further aspect of the subject matter of the present application, there is also provided a rotary cutting tool comprising: [0025] a replaceable cutting head of the type described above; [0026] a tool holder of the type described above; and [0027] a fastening screw; wherein: [0028] the male coupling member is located in the female coupling member, with the head abutment surface abutting the holder abutment surface; [0029] each driven projection is located in a respective driving recess, with the driven surface abutting the driving surface; and [0030] the replaceable cutting head is removably attached to the tool holder by the fastening screw located in the head through hole and being threadingly engaged with the holder internal threaded hole.

[0031] It is understood that the above-said is a summary, and that features described hereinafter may be applicable in any combination to the subject matter of the present application, for example, any of the following features may be applicable to the replaceable cutting head, the tool holder or the rotary cutting tool:

[0032] Each projecting wall can further comprise opposite radially outward and inward projection surfaces which connect the rotationally leading trailing projection flank surface. The radially outward and inward projection surfaces can merge smoothly with the conically shaped head abutment surface and the through hole peripheral surface, respectively.

[0033] Each driven surface can be planar.

[0034] The head rearward surface can define a head rearward plane oriented perpendicular to the head central axis. Each planar driven surface can form an external driven axial angle with the head rearward plane. The driven axial angle can fulfill the condition: $100^{\circ} \leq \mu_1 \leq 130^{\circ}$.

[0035] The driven axial angle can fulfill the condition: $\mu_1 = 115^{\circ}$.

[0036] In a cross-sectional view of the replaceable cutting head taken in a head radial plane oriented perpendicular to the head central axis and extending through a given driven surface, the given driven surface can define an imaginary straight driven line which intersects the head through hole.

[0037] The given driven surface can comprise a radially outermost driven point and a radially innermost driven point, the radially outermost driven point being angularly offset from the radially innermost driven point, relative to the head central axis, by a non-zero acute driven offset angle. An imaginary straight driven radial line can extend from the head central axis and intersect the radially outermost driven point, the imaginary straight driven radial line forming a non-zero acute driven radial angle with the given driven surface.

[0038] The head radial plane can extend through a forwardmost portion of the given driven surface. The driven radial angle can fulfill the condition: $15^{\circ} \leq \varepsilon_1 \leq 25^{\circ}$.

[0039] The radially outermost driven point can be angularly offset from the radially innermost driven point, relative to the head central axis, in the rotationally forward direction.

[0040] The head axial abutment surface can be planar and oriented perpendicular to the head central axis. The head axial abutment surface can extend entirely circumferentially about the male coupling member.

[0041] The at least two driven projections can be identical.

[0042] The male coupling member can comprise exactly two angularly spaced apart driven projections.

[0043] Each driven projection can subtend a maximum projection angle at the head central axis. The maximum projection angle can fulfill the condition: $100^{\circ} \leq \theta_1 \leq 130^{\circ}$.

[0044] The head abutment surface can taper inwardly in the rearward direction at a head cone angle. The head cone angle can be in the range of $5.5^{\circ} \leq 2\alpha \leq 6.5^{\circ}$.

[0045] The head cone angle can be in the range of $5.9^{\circ} \leq 2\alpha \leq 6.1^{\circ}$.

[0046] The cutting portion can comprise at least two peripheral cutting edges defining an outer cutting diameter. The outer cutting diameter can fulfill the condition: $6 \text{ mm} < O < 8 \text{ mm}$.

[0047] The driven projections can have a projection height as measured in a direction along the head central axis. The male coupling member can have a male coupling member height as measured between the head rearward surface and the head base surface in the direction along the head central axis. The projection height can fulfill the condition: $0.25H' \leq H \leq 0.50H'$.

[0048] The cutting portion can have a cutting portion height as measured between the head forward surface and the head base surface in the direction along the head central axis. The male coupling member height can be less than the cutting portion height.

[0049] Each recess circumferential wall can further comprise a radially outward recess surface which connects the rotationally leading and trailing recess flank surfaces. The radially outward recess surface can extend to, and merge smoothly with, the conically shaped holder abutment surface. Each driving recess can open out to the holder internal threaded hole at a radially inward recess opening opposite the radially outward recess surface.

[0050] Each driving surface can be planar.

[0051] The female member bottom surface can define a female member bottom plane oriented

perpendicular to the holder longitudinal axis. Each planar driving surface can form an internal driving axial angle with the female member bottom plane. The driving axial angle can fulfill the condition: $100^{\circ} \leq \mu_2 \leq 130^{\circ}$.

[0052] The driving axial angle can fulfill the condition: $\mu_2 = 115^{\circ}$.

[0053] In a cross-sectional view of the tool holder taken in a holder radial plane oriented perpendicular to the holder longitudinal axis and extending through a given driving surface, the given driving surface can define an imaginary straight driving line which intersects the holder internal threaded hole.

[0054] The given driving surface can comprise a radially outermost driving point and a radially innermost driving point, the radially outermost driving point being angularly offset from the radially innermost driving point, relative to the holder longitudinal axis, by a non-zero acute driving offset angle. An imaginary straight driving radial line can extend from the head central axis and intersect the radially outermost driving point, the imaginary straight driving radial line forming a non-zero acute driving radial angle with the given driving surface.

[0055] The holder radial plane can extend through a forwardmost portion of the given driving surface. The driving radial angle can fulfill the condition: $15^{\circ} \leq \varepsilon_2 \leq 25^{\circ}$.

[0056] The radially outermost driving point can be angularly offset from the radially innermost driving point, relative to the holder longitudinal axis, in the rotationally forward direction.

[0057] The holder axial abutment surface can be planar and oriented perpendicular to the holder longitudinal axis. The holder axial abutment surface can extend entirely circumferentially about the female coupling member.

[0058] The at least two driving recesses can be identical.

[0059] The holder pocket can comprise exactly two angularly spaced apart driving recesses.

[0060] Each driving recess can subtend a maximum recess angle at the holder longitudinal axis. The recess angle can fulfill the condition: $100^{\circ} \leq \theta_2 \leq 130^{\circ}$.

[0061] The holder abutment surface can taper inwardly in the rearward direction, at a holder cone angle. The holder cone angle can be in the range of $5.5^{\circ} \leq 2\gamma \leq 6.5^{\circ}$.

[0062] The holder cone angle can be in the range of $5.9^{\circ} \leq 2\gamma \leq 6.1^{\circ}$.

[0063] Each driving recess comprises a recess bottom surface bounded by the recess circumferential wall.

[0064] The driving recesses can have a recess depth as measured in a direction along the holder longitudinal axis. The female coupling member can have a female coupling member depth as measured between the holder forward surface and the female member bottom surface in the direction along the holder longitudinal axis. The recess depth can fulfill the condition: $0.25D' \leq D \leq 0.50D'$.

[0065] In a locked position of the cutting tool, the head axial abutment surface can abut the holder axial abutment surface.

[0066] In a locked position of the cutting tool, each rotationally leading projection flank surface can be spaced apart from a respective rotationally leading recess flank surface.

[0067] In a locked position of the cutting tool, the head rearward surface can be spaced apart from the female member bottom surface.

[0068] The rotary cutting tool can be a reamer.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0069] For a better understanding of the present application and to show how the same may be carried out in practice, reference will now be made to the accompanying drawings, in which:

[0070] FIG. 1 is a perspective view of a rotary cutting tool;

[0071] FIG. 2 is an exploded perspective view of the rotary cutting tool shown in FIG. 1;
[0072] FIG. 3 is a perspective view of a replaceable cutting head shown in FIGS. 1 and 2;
[0073] FIG. 4 is a side view of the replaceable cutting head shown in FIG. 3;
[0074] FIG. 5 is another side view of the replaceable cutting head shown in FIG. 3;
[0075] FIG. 6 is a cross-sectional view of the replaceable cutting head taken along the line VI-VI in FIG. 5;
[0076] FIG. 6A is detail of FIG. 6;
[0077] FIG. 7 is a rearward view of the replaceable cutting head shown in FIG. 3;
[0078] FIG. 8 is a cross-sectional view of the replaceable cutting head taken along the line VIII-VIII in FIG. 7;
[0079] FIG. 9 is a forward view of the replaceable cutting head shown in FIG. 3;
[0080] FIG. 10 is a perspective view of a forward end of a tool holder shown in FIGS. 1 and 2, in front of a holder pocket;
[0081] FIG. 11 is a forward view of the tool holder shown in FIG. 10;
[0082] FIG. 12 is a cross-sectional view of the tool holder taken along the line XII-XII in FIG. 11;
[0083] FIG. 13 is a side view of the tool holder shown in FIG. 10;
[0084] FIG. 14 is a cross-sectional view of the tool holder taken along the line XIV-XIV in FIG. 13;
[0085] FIG. 14A is detail of FIG. 14;
[0086] FIG. 15 is a forward view of the rotary cutting tool shown in FIGS. 1 and 2;
[0087] FIG. 16 is a cross-sectional view of the rotary cutting tool taken along the line XVI-XVI in FIG. 15;
[0088] FIG. 17 is a side view of the rotary cutting tool shown in FIG. 15; and
[0089] FIG. 18 is a cross-sectional view of the rotary cutting tool taken along the line XVIII-XVIII in FIG. 17.

[0090] It will be appreciated that for simplicity and clarity of illustration, elements shown in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity, or several physical components may be included in one functional block or element. Where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements.

DETAILED DESCRIPTION OF THE INVENTION

[0091] In the following description, various aspects of the subject matter of the present application will be described. For purposes of explanation, specific configurations and details are set forth in sufficient detail to provide a thorough understanding of the subject matter of the present application. However, it will also be apparent to one skilled in the art that the subject matter of the present application can be practiced without the specific configurations and details presented herein.

[0092] Attention is first drawn to FIGS. 1 and 2 showing a rotary cutting tool 20 of the type used for reaming operations, in accordance with embodiments of the subject matter of the present application. The rotary cutting tool 20 includes a replaceable cutting head 22 that has a head central axis A, around which the replaceable cutting head 22 rotates defining opposite rotationally forward and rearward directions R.sub.F, R.sub.R. The rotationally forward direction R.sub.F is the cutting direction. The head central axis A also defines opposite forward and rearward directions D.sub.F, D.sub.R. Stated differently, the head central axis A extends in a forward D.sub.F to rearward direction D.sub.R. The replaceable cutting head 22 can be typically made from cemented carbide. The rotary cutting tool 20 also includes a tool holder 24. The tool holder 24 can be typically made from steel. The replaceable cutting head 22 can be removably retained in the tool holder 24 by means of a tapered coupling mechanism and a fastening screw 25. Such couplings could possibly be advantageous for other types of rotary cutting operations than that stated hereinabove, such as, for example, milling or drilling.

[0093] It should be appreciated that use of the terms “forward” and “rearward” throughout the description and claims when referring to the cutting head **22** relate to a relative position in a direction of the head central axis A towards the top and bottom of the page, respectively, in FIGS. **4** and **5**. Likewise, when referring to the tool holder **24** the two terms relate to a relative position in a direction of the holder longitudinal axis C towards the left and right of the page, respectively, in FIGS. **12** and **13**. It should be further appreciated that use of the terms “rotationally forward” and “rotationally rearward” throughout the description and claims refer to a relative position in a rotational direction about the head central axis A counter-clockwise and clockwise, respectively, in FIG. **9**. Likewise, when referring to the tool holder **24** the two terms relate to a relative position in a rotational direction about the holder longitudinal axis C counterclockwise and clockwise, respectively, in FIG. **14**.

[0094] Reference is now made to FIGS. **3** to **9**. The replaceable cutting head **22** has a forward portion that forms a cutting portion **26** and a rearward portion that forms a mounting portion **28**. That is to say, the cutting portion **26** is forwardly disposed and the mounting portion **28** is rearwardly disposed. In accordance with some embodiments of the subject matter of the present application, the cutting portion **26** can adjoin the mounting portion **28**.

[0095] Referring to FIG. **4**, the replaceable cutting head **22** includes opposite head forward and rearward surfaces **48**, **50** and a head peripheral surface **52** extending therebetween. The head forward surface **48** is located at the cutting portion **26** and faces generally in the forward direction D.sub.F. The head rearward surface **50** is located at the mounting portion **28** and faces generally in the rearward direction D.sub.R. The head peripheral surface **52** extends about the head central axis A. In accordance with some embodiments of the subject matter of the present application, the head rearward surface **50** can define a head rearward plane HP1 oriented perpendicular to the head central axis A. The head rearward surface **50** can lie in the head rearward plane HP1.

[0096] In accordance with some embodiments of the subject matter of the present application the replaceable cutting head **22** can be integrally formed to have unitary one-piece construction. This provides an advantage in that the replaceable cutting head **22** has no detachable cutting inserts (not shown). Such detachable cutting inserts need to be replaced periodically and this can be a time-consuming procedure. There is also a possibility that threaded screws (not shown), for example, which can be used to releasably retain the detachable cutting inserts to the replaceable cutting head **22** can be mislaid and/or lost during the replacement operation.

[0097] Referring to FIGS. **6** and **7**, the cutting portion **26** includes at least one peripheral cutting edge **30**. In this non-limiting example shown in the drawings, there can be exactly six peripheral cutting edges. The peripheral cutting edges **30** define an outer cutting diameter O. The outer cutting diameter O can be greater than or equal to 6 mm and less than or equal to 8 mm. The outer cutting diameter O can exactly equal to 6 mm. Each peripheral cutting edge **30** is formed at the intersection of a peripheral relief surface **32**, and a peripheral rake surface **34**. The peripheral relief surface **32** is located rotationally rearward of (i.e., behind) the peripheral cutting edge **30** and the peripheral rake surface **34** is located rotationally forward of (i.e., ahead) of the peripheral cutting edge **30**.

Generally speaking, the peripheral rake surface **34** faces in the rotationally forward direction R.sub.F. The orientation of the peripheral cutting edge **30** (and the peripheral rake and relief surfaces **34**, **32**) allows metal cutting operations to be performed. In accordance with some embodiments of the subject matter of the present application the cutting portion **26** can include at least one flute **36** for evacuating chips (not shown) that are produced during the cutting operation. One flute **36** is associated with each peripheral cutting edge **30**. As seen in the figures, the cutting portion **26** includes a plurality circumferentially arranged cutting sections **31**, each cutting section **31** including in the rotationally forward direction R.sub.F, a trailing relief surface **32**, a cutting edge **30**, a rake surface **34** and a flute **36**, one after another.

[0098] Making reference in particular to FIG. **4**, the mounting portion **28** includes a male coupling member **38** that protrudes rearwardly from a head base surface **40**. The head base surface **40**

extends transversely with respect to the head central axis A and defines a boundary between the cutting portion **26** and the mounting portion **28**. That is to say, the cutting portion **26** is formed forward of the head base surface **40** and the mounting portion **28** is formed rearward of the head base surface **40**. In accordance with some embodiments of the subject matter of the present application the male coupling member **38** can be rigid. It is noted that the male coupling member **38** can be delimited in the rearward direction D.sub.R by the head rearward surface **50**.

[0099] The head base surface **40** can include a head axial abutment surface **41**, for abutment with a corresponding surface on the tool holder **24**, thereby allowing precise axial seating of the cutting head **22** in the tool holder **24**. In accordance with some embodiments of the subject matter of the present application, the head axial abutment surface **41** can be planar and oriented perpendicular to the head central axis A. Referring to FIG. 3, the head axial abutment surface **41** can extend entirely circumferentially about the male coupling member **38**.

[0100] The male coupling member **38** includes a male member peripheral surface **39** which extends between the head base surface **40** and the head rearward surface **50**. The male member peripheral surface **39** extends about the head central axis A.

[0101] Referring again to FIG. 4, the cutting portion **26** has a cutting portion height H'' as measured between the head forward surface **48** and the head base surface **40** in a direction along the head central axis A. The male coupling member **38** has a male coupling member height H' as measured between the head rearward surface **50** and the head base surface **40** in the direction along the head central axis A. In accordance with some embodiments of the subject matter of the present application, the male coupling member height H' can be less than the cutting portion height H''.

[0102] Referring to FIG. 4, the male coupling member **38** includes a circumferentially extending head abutment surface **46** that extends about the head central axis A. The head abutment surface **46** is located on the male member peripheral surface **39**. The head abutment surface **46** faces radially outwardly. The head abutment surface **46** tapers inwardly in the rearward direction D.sub.R at a head cone angle 2α . That is to say, the head abutment surface **46** has a non-cylindrical, conical shape facing radially outwards, where the head cone angle 2α is an internal angle. In accordance with some embodiments of the subject matter of the present application, the head abutment surface **46** can be frusto-conical. The head cone angle 2α can be in the range of $5.5^\circ \leq 2\alpha \leq 6.5^\circ$. Stated differently, the head abutment surface **46** can define a taper angle α which is in the range of $2.75^\circ \leq \alpha \leq 3.25^\circ$ with respect to the head central axis A (the taper angle α being half the cone angle 2α). Preferably, the head cone angle 2α can be in the range of $5.9^\circ \leq 2\alpha \leq 6.1^\circ$. Further preferably, the head cone angle 2α can equal 6° . It is noted that the head abutment surface **46** is intended to abut a corresponding surface on the tool holder **24** when the rotary cutting tool **20** is in a locked position, as will be described hereinafter. The conically shaped head abutment surface **46** acts to center the cutting head **22** in the tool holder **24**. The head abutment surface **46** can extend axially between the head base surface **40** and the head rearward surface **50**.

[0103] It should be appreciated that use of the terms “radially inward/inwardly” and “radially outward/outwardly” throughout the description and claims refer to a relative position in a perpendicular direction in relation to the head central axis A and/or holder longitudinal axis C, towards and away from the respective axis, in FIGS. 4 to 9, and 10 to 12. It should further be appreciated that use of the term “cone angle” throughout the description refers to an angle formed by the tapered surfaces of a cone, in a longitudinal cross-section. It is noted that the term “longitudinal cross-section” refers to a cross-section taken in a plane containing the longitudinal axis.

[0104] Referring in general to FIGS. 2-9, the replaceable cutting head **22** includes a head through hole **42**, which extends through the cutting and mounting portions **26**, **28**. The head through hole **42** is configured for the fastening screw **25** to be inserted therethrough, as described later in the description. The head through hole **42** includes a through hole peripheral surface **44** which extends about a through hole central axis E. Generally speaking, the through hole peripheral surface **44**

faces towards the through hole central axis E. The through hole peripheral surface **44** intersects the head forward and rearward surfaces **48**, **50**. That is to say, the head through hole **42** opens out to the head forward and rearward surfaces **48**, **50** (at head forward and rearward hole openings **49**, **51**, respectively). In accordance with some embodiments of the subject matter of the present application, the head forward and rearward hole openings **49**, **51** can be intersected by the head central axis A. The through hole central axis E can be co-incident with the head central axis A. It is noted that the head through hole **42** can be devoid of an internal thread.

[0105] As seen in FIGS. **3** and **7**, the mounting portion **28** includes at least two angularly spaced apart driven projections **54** which project from the head rearward surface **50**. In accordance with some embodiments of the subject matter of the present application, the at least two driven projections **54** can be identical. In some embodiments, the male coupling member **38** can include exactly two angularly spaced apart driven projections **54**.

[0106] The following disclosure refers to one of the driven projections **54** but is applicable to all other driven projections **54**.

[0107] Referring to FIG. **4**, the driven projection **54** has a projection height H, as measured in the direction along the head central axis A (between the head rearward surface **50** and a projection top surface **58**, described below). The projection height H can be greater than or equal to 25% of the male coupling member height H' and less than or equal to 50% of the male coupling member height H' (i.e., $0.25H' \leq H \leq 0.50H'$).

[0108] Referring to FIGS. **6A** and **7**, the driven projection **54** includes a projecting wall **56** and the projection top surface **58** bounded by the projecting wall **56**. The projecting wall **56** includes two opposite and angularly spaced apart projection flank surfaces **60L**, **60T**. That is, the two opposite projection flank surfaces **60L**, **60T** include a rotationally leading and a rotationally trailing projection flank surface **60L**, **60T** (the rotationally leading projection flank surface **60L** being rotationally forward of the rotationally trailing projection flank surface **60T**). The projecting wall **56** includes opposite radially inward and outward projection surfaces **62I**, **62O** which connect the rotationally leading and trailing projection flank surfaces **60L**, **60T**. In accordance with some embodiments of the subject matter of the present application, the radially outward projection surface **62O** can merge smoothly with the conically shaped head abutment surface **46**. For instance, the radially outward projection surface **62O** may lay on the same imaginary conical surface as the conically shaped head abutment surface **46**. Likewise, the radially inward projection surface **62I** can merge smoothly with the through hole peripheral surface **44**. Thus, the inward projection surface **62I** may lay on the same imaginary cylindrical surface as the through hole peripheral surface **44**, or even be considered an extension of the latter.

[0109] The rotationally trailing projection flank surface **60T** includes a driven surface **64**, for opposing a torque about the head central axis A. In accordance with some embodiments of the subject matter of the present application, the driven surface **64** can be planar. A planar driven surface **64** may be more efficient at torque transfer than a convexly curved driven surface due to more complete contact with a complementary driving surface. As seen in the figures, driven surfaces **64** are located at the rearwardmost portion of the cutting head **22** and constitute the only torque-receiving surfaces formed on the cutting head **22**. As such, no driven surfaces are present in the cutting portion **26** of the cutting head **22**. Furthermore, the entire cutting head **22** with its driven surfaces **64** has monolithic (unitary, one-piece) construction.

[0110] Reverting to FIG. **4**, in accordance with some embodiments of the subject matter of the present application, the planar driven surface **64** can form an external driven axial angle μ_1 with the head rearward plane HP1. The driven axial angle μ_1 can fulfill the condition: $100^\circ \leq \mu_1 \leq 130^\circ$. Preferably, the driven axial angle μ_1 can fulfill the condition: $110^\circ \leq \mu_1 \leq 120^\circ$. Further preferably, the driven axial angle μ_1 can fulfill the condition: $\mu_1 = 115^\circ$. A drive force F exerted by a driving surface **94** on each driving recess **80** (discussed further below) acts on the driven surface **64** on a respective driven projection **54** (in a direction normal to the driven surface **64**). In extreme

conditions said drive force F can cause the driven projection **54** to break (shear off). Forming the driven axial angle $\mu 1$ to be in the range of $100^{\circ} \leq \mu 1 \leq 130^{\circ}$ has been found to advantageously reduce the possibility of breakage by directing the drive force F upwards towards the male coupling member **38**, such that the driven projection **54** is supported/strengthened by the male coupling member **38** (against said drive force F). Moreover, having the driven axial angle $\mu 1$ less than 130° ensures that the driven surface **64** does not slip against the driving recess **80**. It should be appreciated that throughout the detailed description and claims, an “internal angle” refers to an angle between two surface components of a member surface as measured internal to the member, whereas an “external angle” refers to an angle between two surface components of a member surface as measured external to the member.

[0111] Reverting to FIGS. **6** and **6A**, in accordance with some embodiments of the subject matter of the present application, in a cross-sectional view of the replaceable cutting head **22** taken in a head radial plane RP1 oriented perpendicular to the head central axis A and extending through a given driven surface **64**, the given driven surface **64** defines an imaginary straight driven line DL1 which may intersect the head through hole **42**.

[0112] In accordance with some embodiments of the subject matter of the present application, the given driven surface **64** can include a radially outermost driven point PO1 and a radially innermost driven point PI1. The radially outermost driven point PO1 can be angularly offset from the radially innermost driven point PI1, relative to the head central axis A, by a non-zero acute driven offset angle $\beta 1$. Thus, the straight driven line DL1 may not intersect the head central axis A. The radially outermost driven point PO1 can be angularly offset, relative to the head central axis A, from the radially innermost driven point PI1 in the rotationally forward direction R.sub.F. The driven offset angle $\beta 1$ can fulfill the condition: $10^{\circ} \leq \beta 1 \leq 15^{\circ}$. Forming the driven offset angle $\beta 1$ to be in this range has been found advantageous for strengthening a radially outer corner portion **650** (formed at the intersection of the radially outward projection surface **62O** and the rotationally trailing projection flank surface **60T**) of each driven projection **54** against the drive force F, while maintaining sufficient strength at a radially inner corner portion **65I** (formed at the intersection of the radially inward projection surface **62I** and the rotationally trailing projection flank surface **60T**) of the same driven projection **54** (see FIG. **6**). Additionally, the contact area between the driven projection **54** and the driving recess **80** is increased (compared to when $\beta 1 = 0$), which is particularly desirable for small diameter tools where contact region is correspondingly small.

[0113] An imaginary straight driven radial line RL1 extends from the head central axis A and intersects the radially outermost driven point PO1. In accordance with some embodiments of the subject matter of the present application, the imaginary straight driven radial line RL1 forms a non-zero acute driven radial angle $\epsilon 1$ with the given driven surface **64**. The head radial plane RP1 can extend through a forwardmost portion of the given driven surface **64** and the driven radial angle $\epsilon 1$ can fulfill the condition: $15^{\circ} \leq \epsilon 1 \leq 25^{\circ}$.

[0114] Making reference to FIG. **7**, each driven projection **54** subtends a maximum projection angle $\theta 1$ at the head central axis A. In accordance with some embodiments of the subject matter of the present application, the maximum projection angle $\theta 1$ can fulfill the condition: $100^{\circ} \leq \theta 1 \leq 130^{\circ}$. In this non-limiting example shown in the drawings, the maximum projection angle $\theta 1$ is defined by the forwardmost portions of the two angularly spaced apart projection flank surfaces **60L**, **60T**.

[0115] Another aspect of the subject matter of the present application relates to the tool holder **24**. Referring now to FIGS. **10-14A**, the tool holder **24** has a holder longitudinal axis C, around which the tool holder **24** rotates defining opposite rotationally forward and rearward directions R.sub.F, R.sub.R. The rotationally forward direction R.sub.F is the cutting direction. The holder longitudinal axis C also defines opposite forward and rearward directions D.sub.F, D.sub.R. In accordance with some embodiments of the subject matter of the present application, the tool holder **24** can include a holder peripheral surface **74** which extends about the holder longitudinal axis C and delimits a boundary of a holder forward surface **70**. The holder forward surface **70** is located at a forward end

of the tool holder **24**. The holder forward surface **70** extends transversely with respect to the holder longitudinal axis C. In accordance with some embodiments of the subject matter of the present application, the holder forward surface **70** can be perpendicular to the holder longitudinal axis C. [0116] In accordance with some embodiments of the subject matter of the present application, the holder peripheral surface **74** can include a forward holder peripheral surface **76** and a rearward holder peripheral surface **78** and an intermediate holder peripheral surface **77** which extends therebetween. The forward holder peripheral surface **76** can be closer to the holder forward surface **70** than both the intermediate holder peripheral surface **77** and the rearward holder peripheral surface **78**. The forward holder peripheral surface **76** and the rearward holder peripheral surface **78** lie on different imaginary co-axial cylinders. The forward holder peripheral surface **76** can have a diameter less than the diameter of the rearward holder peripheral surface **78**. The intermediate holder peripheral surface **77** can slope radially outwardly from the forward holder peripheral surface **76** to the rearward holder peripheral surface **78**.

[0117] The holder forward surface **70** can include a holder axial abutment surface **71**, for abutment with a corresponding surface on the replaceable cutting head **22**. In accordance with some embodiments of the subject matter of the present application, the holder axial abutment surface **71** can be planar and oriented perpendicular to the holder longitudinal axis C.

[0118] Referring to FIG. **12**, the tool holder **24** includes a holder pocket **79** which extends rearwardly from the holder forward surface **70**.

[0119] The holder pocket **79** includes a female coupling member **66** that is recessed in the holder forward surface **70**. The holder axial abutment surface **71** can extend entirely circumferentially about the female coupling member **66**.

[0120] The female coupling member **66** includes a female member bottom surface **68** and a female member peripheral surface **69** extending about the holder longitudinal axis C between the female member bottom surface **68** and the holder forward surface **70**. The female member bottom surface **68** defines a female member bottom plane FP2 oriented perpendicular to the holder longitudinal axis C. The female member bottom surface **68** can lie in the female member bottom plane FP2.

[0121] Referring again to FIG. **12**, the female coupling member **66** has a female coupling member depth D' as measured between the holder forward surface **70** and the female member bottom surface **68** in the direction along the holder longitudinal axis C.

[0122] The female coupling member **66** includes a holder abutment surface **71A** that extends about the holder longitudinal axis C. The holder abutment surface **71A** is located on the female member peripheral surface **69**. The holder abutment surface **71A** faces inwardly with respect to the holder longitudinal axis C. The holder abutment surface **71A** tapers inwardly in the rearward direction D.sub.R at a holder cone angle 2γ . That is to say, the holder abutment surface **71A** has a non-cylindrical, conical shape facing radially inwards, where the holder cone angle 2γ is an external angle. In accordance with some embodiments of the subject matter of the present application, the holder abutment surface **71A** can be frusto-conical. The holder cone angle 2γ can be in the range of $5.5^\circ \leq 2\gamma \leq 6.5^\circ$. Stated differently, the holder abutment surface **71A** can define a taper angle γ which is in the range of $2.75^\circ \leq \gamma < 3.25^\circ$ with respect to the holder longitudinal axis C. Preferably, the holder cone angle 2γ can be in the range of $5.9^\circ \leq 2\gamma \leq 6.1^\circ$. Further preferably, the holder cone angle 2γ can equal 6° . The holder abutment surface **71A** can extend between the holder forward surface **70** and the female member bottom surface **68**.

[0123] The holder pocket **79** includes a holder internal threaded hole **72** recessed in the female member bottom surface **68** and extending along the holder longitudinal axis C. The holder internal threaded hole **72** serves to threadingly receive the fastening screw **25** as described later in the description. As seen in, e.g., FIG. **12**, the holder internal threaded hole **72** is a blind hole, and thus does not axially extend through the body of the tool holder to the latter's rear end.

[0124] Reverting to FIG. **10**, the holder pocket **79** includes at least two angularly spaced apart driving recesses **80** which are recessed in the female member bottom surface **68**. The number of

driving recesses **80** matches the number of the driven projections **54**. The at least two angularly spaced apart driving recesses **80** are designed for the at least two driven projections **54** to be inserted therein. In accordance with some embodiments of the subject matter of the present application, the at least two driving recesses **80** can be identical. In some embodiments, the holder pocket **79** can include exactly two angularly spaced apart driving recesses **80**.

[0125] The following disclosure refers to one of the driving recesses **80** but is applicable to all other driving recesses **80**.

[0126] The driving recess **80** has a recess depth D (see FIG. **12**), as measured in the direction along the holder longitudinal axis C (between the female member bottom surface **68** and a recess bottom surface **84**, described below). The recess depth D can be greater than or equal to 25% of the female coupling member depth D' and less than or equal to 50% of the female coupling member depth D' (i.e., $0.25D' \leq D \leq 0.50D'$).

[0127] As shown in FIGS. **14** and **14A**, the driving recess **80** includes a recess circumferential wall **82** and a recess bottom surface **84** bounded by the recess circumferential wall **82**. The recess circumferential wall **82** includes two opposite and angularly spaced apart recess flank surfaces **88L**, **88T**. That is, the two opposite recess flank surfaces **88L**, **88T** include a rotationally leading and a rotationally trailing recess flank surface **88L**, **88T** (the rotationally leading recess flank surface **88L** being rotationally forward of the rotationally trailing recess flank surface **88T**). In accordance with some embodiments of the subject matter of the present application, the recess circumferential wall **82** can also include a radially outward recess surface **90O** which connects the rotationally leading and trailing recess flank surfaces **88L**, **88T**. The radially outward recess surface **90O** can merge smoothly with the conically shaped holder abutment surface **71A**. The driving recess **80** can open out to the holder internal threaded hole **72** at a radially inward recess opening **92** opposite the radially outward recess surface **90O**.

[0128] The rotationally trailing recess flank surface **88T** includes a driving surface **94**, for providing a torque about the holder longitudinal axis C (to the driven surface **64**). In accordance with some embodiments of the subject matter of the present application, the driving surface **94** can be planar.

[0129] Making reference to FIG. **10**, in accordance with some embodiments of the subject matter of the present application, the planar driving surface **94** can form an internal driving axial angle μ_2 with the female member bottom plane FP_2 . The driving axial angle μ_2 can fulfill the condition: $100^\circ \leq \mu_2 \leq 130^\circ$. Preferably, the driving axial angle μ_2 can fulfill the condition: $110^\circ \leq \mu_2 \leq 120^\circ$. Further preferably, the driving axial angle μ_2 can fulfill the condition: $\mu_2 = 115^\circ$.

[0130] Reverting to FIGS. **14** and **14A**, in accordance with some embodiments of the subject matter of the present application, in a cross-sectional view of the tool holder **24** taken in a holder radial plane RP_2 oriented perpendicular to the holder longitudinal axis C and extending through a given driving surface **94**, the given driving surface **94** defines an imaginary straight driving line DL_2 which may intersect the holder internal threaded hole **72**.

[0131] In accordance with some embodiments of the subject matter of the present application, the given driving surface **94** can include a radially outermost driving point PO_2 and a radially innermost driving point PI_2 . The radially outermost driving point PO_2 can be angularly offset from the radially innermost driving point PI_2 , relative to the holder longitudinal axis C , by a non-zero acute driving offset angle β_2 . Thus, the straight driving line DL_2 may not intersect the holder longitudinal axis C . The radially outermost driving point PO_2 can be angularly offset from the radially innermost driving point PI_2 , relative to the holder longitudinal axis C , in the rotationally forward direction $R.sub.F$. The driving offset angle β_2 can fulfill the condition: $10^\circ \leq \beta_2 \leq 15^\circ$.

[0132] An imaginary straight driving radial line RL_2 extends from the holder longitudinal axis C and intersects the radially outermost driving point PO_2 . In accordance with some embodiments of the subject matter of the present application, the imaginary straight driving radial line RL_2 forms a non-zero acute driving radial angle ϵ_2 with the given driving surface **94**. The holder radial plane

RP2 can extend through a forwardmost portion of the given driving surface **94** and the driving radial angle ϵ_2 can fulfill the condition: $15^\circ \leq \epsilon_2 \leq 25^\circ$.

[0133] Making reference to FIG. **11**, each driving recess **80** subtends a maximum recess angle θ_2 at the holder longitudinal axis C. In accordance with some embodiments of the subject matter of the present application, the maximum recess angle θ_2 can fulfill the condition: $100^\circ \leq \theta_2 \leq 130^\circ$. In this non-limiting example shown in the drawings, the maximum recess angle θ_2 is defined by the forwardmost portions of the two angularly spaced apart recess flank surfaces **88L**, **88T**.

[0134] Referring to FIGS. **15-18**, another aspect of the subject matter of the present application relates to a rotary cutting tool **20** that includes the replaceable cutting head **22** and tool holder **24** as defined herein above releasably attached by the fastening screw **25**.

[0135] The rotary cutting tool **20** is adjustable between a released position and a locked position. In the released position of the rotary cutting tool **20**, as shown in FIG. **2**, the rotary cutting tool **20** is unassembled and the male coupling member **38** is located outside of the female coupling member **66**.

[0136] Assembly of the rotary cutting tool **20** can be accomplished by performing the following steps. The mounting portion **28** of the replaceable cutting head **22** is placed in front of the holder pocket **79** of the tool holder **24**. Next, the male coupling member **38** is inserted into the female coupling member **66** until each driven projection **54** initially enters a respective driving recess **80** (it may be required to rotate the replaceable cutting head **22** with respect to the tool holder **24** so that the at least two driven projections **54** are rotationally aligned with the at least two driving recesses **80**). Then, the replaceable cutting head **22** is moved further towards tool holder **24** until the head abutment surface **46** comes into circumferential contact with the holder abutment surface **71A**. Next, the fastening screw **25** is inserted through the head through hole **42** and initially threadingly engages with the holder internal threaded hole **72**. Tightening of the fastening screw **25** causes the fastening screw **25** to urge the male coupling member **38** of the replaceable cutting head **22** further into the female coupling member **66** of the tool holder **24** until the conical head abutment surface **46** firmly abuts the conical forward holder abutment surface **71A**, defining the locked position of the rotary cutting tool **22**. The contact between the conical head abutment surface **46** and the conical forward holder abutment surface **71A** can form a friction-fit. In the locked position of the rotary cutting tool **20**, the replaceable cutting head **22** is removably attached to the tool holder **24** by the fastening screw **25** located in the head through hole **42** and being threadingly engaged with the holder internal threaded hole **72**. The male coupling member **38** is located in the female coupling member **66** by the fastening screw **25** located in the head through hole **42**. The fastening screw **25** and the holder internal threaded hole **72** are threadingly engaged with each other. The head central axis A and the holder longitudinal axis C are both co-incident (i.e., aligned) with each other. Each driven projection **54** is located in a respective driving recess **80**, with the driven surface **64** abutting the driving surface **94**. In accordance with some embodiments of the subject matter of the present application, the head axial abutment surface **41** can abut the holder axial abutment surface **71**. Each projection top surface **58** can be spaced apart from a respective recess bottom surface **84**. Each rotationally leading projection flank surface **60L** can be spaced apart from a respective rotationally leading recess flank surface **88L**. The head rearward surface **50** can be spaced apart from the female member bottom surface **68**.

[0137] It should further be noted that a feature of the subject matter of the present application is that the torque transfer system (i.e., the location of driving and driven surfaces) is compact. For example, the torque transfer system is located centrally and in particular radially within the centering mechanism (i.e., the conically shaped head and holder abutment surfaces).

[0138] Although the subject matter of the present application has been described to a certain degree of particularity, it should be understood that various alterations and modifications could be made without departing from the spirit or scope of the invention as hereinafter claimed.

Claims

1. A replaceable cutting head (22), for rotary cutting operations, having a head central axis (A), defining opposite forward and rearward directions (D.sub.F, D.sub.R), and opposite rotationally forward and rearward directions (R.sub.F, R.sub.R) with the rotationally forward direction (R.sub.F) being the cutting direction, the replaceable cutting head (22) comprising: a forward portion forming a cutting portion (26) and a rearward portion forming a mounting portion (28); and opposite head forward and rearward surfaces (48, 50) and a head peripheral surface (52) extending therebetween, the head forward surface (48) being located at the cutting portion (26) and the head rear surface (50) being located at the mounting portion (28); a head through hole (42) comprising a through hole peripheral surface (44) which extends about a through hole central axis (E), and intersects the head forward and rearward surfaces (48, 50); wherein: the mounting portion (28) comprises: a male coupling member (38) protruding rearwardly from a head base surface (40), the head base surface (40) extending transversely with respect to the head central axis (A), and defining a boundary between the cutting portion (26) and the mounting portion (28); and at least two angularly spaced apart driven projections (54) projecting from the head rearward surface (50), each driven projection (54) comprising a projecting wall (56) comprising two opposite and angularly spaced apart projection flank surfaces (60L, 60T) including a rotationally leading and a rotationally trailing projection flank surface (60L, 60T), the rotationally trailing projection flank surface (60T) comprising a driven surface (64), for opposing a torque about the head central axis (A); wherein: the head base surface (40) comprises a head axial abutment surface (41); and the male coupling member (38) comprises a non-cylindrical, conically shaped head abutment surface (46).
2. The replaceable cutting head (22) according to claim 1, wherein: each projecting wall (56) further comprises opposite radially outward and inward projection surfaces (62O, 62I) which connect the rotationally leading trailing projection flank surface (60L, 60T); and the radially outward and inward projection surfaces (62O, 62I) merge smoothly with the conically shaped head abutment surface (46) and the through hole peripheral surface (44), respectively.
3. The replaceable cutting head (22) according to claim 1, wherein: each driven surface (64) is planar.
4. The replaceable cutting head (22) according to claim 3, wherein: the head rearward surface (50) defines a head rearward plane (HP1) oriented perpendicular to the head central axis (A); each planar driven surface (64) forms an external driven axial angle (μ_1) with the head rearward plane (HP1); and the driven axial angle (μ_1) fulfills the condition: $100^\circ \leq \mu_1 \leq 130^\circ$.
5. The replaceable cutting head (22) according to claim 4, wherein: the driven axial angle (μ_1) fulfills the condition: $\mu_1 = 115^\circ$.
6. The replaceable cutting head (22) according to claim 1, wherein: in a cross-sectional view of the replaceable cutting head (22) taken in a head radial plane (RP1) oriented perpendicular to the head central axis (A) and extending through a given driven surface (64): the given driven surface (64) defines an imaginary straight driven line (DL1) which intersects the head through hole (42).
7. The replaceable cutting head (22) according to claim 6, wherein: the given driven surface (64) comprises a radially outermost driven point (PO1) and a radially innermost driven point (PI1), the radially outermost driven point (PO1) being angularly offset from the radially innermost driven point (PI1), relative to the head central axis (A), by a non-zero acute driven offset angle (β_1); an imaginary straight driven radial line (RL1) extends from the head central axis (A) and intersects the radially outermost driven point (PO1), the imaginary straight driven radial line (RL1) forming a non-zero acute driven radial angle (ϵ_1) with the given driven surface (64).
8. The replaceable cutting head (22) according to claim 7, wherein: the head radial plane (RP1) extends through a forwardmost portion of the given driven surface (64); and the driven radial angle

(81) fulfills the condition: $15^{\circ} \leq \epsilon_1 \leq 25^{\circ}$.

9. The replaceable cutting head (22) according to claim 7, wherein: the radially outermost driven point (PO1) is angularly offset from the radially innermost driven point (PI1), relative to the head central axis (A), in the rotationally forward direction (R.sub.F).

10. The replaceable cutting head (22) according to claim 1, wherein: the head axial abutment surface (41) is planar and oriented perpendicular to the head central axis (A); and the head axial abutment surface (41) extends entirely circumferentially about the male coupling member (38).

11. The replaceable cutting head (22) according to claim 1, wherein: the at least two driven projections (54) are identical.

12. The replaceable cutting head (22) according to claim 1, wherein: the male coupling member (38) comprises exactly two angularly spaced apart driven projections (54).

13. The replaceable cutting head (22) according to claim 1, wherein: each driven projection (54) subtends a maximum projection angle (θ_1) at the head central axis (A); and the maximum projection angle (θ_1) fulfills the condition: $100^{\circ} \leq \theta_1 \leq 130^{\circ}$.

14. The replaceable cutting head (22) according to claim 1, wherein: the cutting portion (26) comprises at least two peripheral cutting edges (30) defining an outer cutting diameter (O); and the outer cutting diameter (O) fulfills the condition: $6 \text{ mm} < O < 8 \text{ mm}$.

15. The replaceable cutting head (22) according to claim 1, wherein: the driven projections (54) have a projection height (H) as measured in a direction along the head central axis (A); the male coupling member (38) has a male coupling member height (H') as measured between the head rearward surface (50) and the head base surface (40) in the direction along the head central axis (A); and the projection height (H) fulfills the condition: $0.25H' \leq H \leq 0.50H'$.

16. The replaceable cutting head (22) according to claim 15, wherein: the cutting portion (26) has a cutting portion height (H'') as measured between the head forward surface (48) and the head base surface (40) in the direction along the head central axis (A); and the male coupling member height (H') is less than the cutting portion height (H'').

17. A tool holder (24), having a holder longitudinal axis (C) defining opposite forward and rearward directions (D.sub.F, D.sub.R), and opposite rotationally forward and rearward directions (R.sub.F, R.sub.R) with the rotationally forward direction (R.sub.F) being the cutting direction, the tool holder (24) comprising a holder pocket (79) extending rearwardly from a holder forward surface (70), the holder forward surface (70) extending transversely with respect to the holder longitudinal axis (C); wherein: the holder pocket (79) comprises: a female coupling member (66) recessed in the holder forward surface (70), the female coupling member (66) comprising: a non-cylindrical, conically shaped holder abutment surface (71A); and a female member bottom surface (68); at least two angularly spaced apart driving recesses (80) recessed in the female member bottom surface (68), each driving recess (80) comprising a recess circumferential wall (82) comprising two opposite and angularly spaced apart recess flank surfaces (88L, 88T) including a rotationally leading and a rotationally trailing recess flank surface (88L, 88T), the rotationally trailing recess flank surface (88T) comprising a driving surface (94), for providing a torque about the holder longitudinal axis (C); and a holder internal threaded hole (72) recessed in the female member bottom surface (68); wherein: the holder forward surface (70) comprises a holder axial abutment surface (71).

18. The tool holder (24) according to claim 17, wherein: each recess circumferential wall (82) further comprises a radially outward recess surface (90O) which connects the rotationally leading and trailing recess flank surfaces (88L, 88T); the radially outward recess surface (90O) extends to, and merges smoothly with, the conically shaped holder abutment surface (71A); and each driving recess (80) opens out to the holder internal threaded hole (72) at a radially inward recess opening (92) opposite the radially outward recess surface (90O).

19. The tool holder (24) according to claim 17, wherein: each driving surface (94) is planar.

20. The tool holder (24) according to claim 17, wherein: the female member bottom surface (68)

defines a female member bottom plane (FP2) oriented perpendicular to the holder longitudinal axis (C); each planar driving surface (94) forms an internal driving axial angle (μ_2) with the female member bottom plane (FP2); and the driving axial angle (μ_2) fulfills the condition: $100^\circ \leq \mu_2 \leq 130^\circ$.

21. The tool holder (24) according to claim 20, wherein: the driving axial angle (μ_2) fulfills the condition: $\mu_2 = 115^\circ$.

22. The tool holder (24) according to claim 17, wherein: in a cross-sectional view of the tool holder (24) taken in a holder radial plane (RP2) oriented perpendicular to the holder longitudinal axis (C) and extending through a given driving surface (94): the given driving surface (94) defines an imaginary straight driving line (DL2) which intersects the holder internal threaded hole (72).

23. The tool holder (24) according to claim 22, wherein: the given driving surface (94) comprises a radially outermost driving point (PO2) and a radially innermost driving point (PI2), the radially outermost driving point (PO2) being angularly offset from the radially innermost driving point (PI2), relative to the holder longitudinal axis (C), by a non-zero acute driving offset angle (β_2); an imaginary straight driving radial line (RL2) extends from the head central axis (A) and intersects the radially outermost driving point (PO2), the imaginary straight driving radial line (RL2) forming a non-zero acute driving radial angle (ϵ_2) with the given driving surface (94).

24. The tool holder (24) according to claim 23, wherein: the holder radial plane (RP2) extends through a forwardmost portion of the given driving surface (94); and the driving radial angle (ϵ_2) fulfills the condition: $15^\circ \leq \epsilon_2 \leq 25^\circ$.

25. The tool holder (24) according to claim 23, wherein: the radially outermost driving point (PO2) is angularly offset from the radially innermost driving point (PI2), relative to the holder longitudinal axis (C), in the rotationally forward direction (R.sub.F).

26. The tool holder (24) according to claim 17, wherein: the holder axial abutment surface (71) is planar and oriented perpendicular to the holder longitudinal axis (C); and the holder axial abutment surface (71) extends entirely circumferentially about the female coupling member (66).

27. The tool holder (24) according to claim 17, wherein: the at least two driving recesses (80) are identical.

28. The tool holder (24) according to claim 17, wherein: the holder pocket (79) comprises exactly two angularly spaced apart driving recesses (80).

29. The tool holder (24) according to claim 17, wherein: each driving recess (80) subtends a maximum recess angle (θ_2) at the holder longitudinal axis (C); and the maximum recess angle (θ_2) fulfills the condition: $100^\circ \leq \theta_2 \leq 130^\circ$.

30. The tool holder (24) according to claim 17, wherein: each driving recess (80) comprises a recess bottom surface (84) bounded by the recess circumferential wall (82); the driving recesses (80) have a recess depth (D) as measured in a direction along the holder longitudinal axis (C); the female coupling member (66) has a female coupling member depth (D') as measured between the holder forward surface (70) and the female member bottom surface (68) in the direction along the holder longitudinal axis (C); and the recess depth (D) fulfills the condition: $0.25D' \leq D \leq 0.50D'$.

31. A rotary cutting tool (20) comprising: a replaceable cutting head (22) in accordance with claim 1; a tool holder (24) in accordance with claim 17; and a fastening screw (25); wherein: the male coupling member (38) is located in the female coupling member (66), with the head abutment surface (46) abutting the holder abutment surface (71A); each driven projection (54) is located in a respective driving recess (80), with the driven surface (64) abutting the driving surface (94); and the replaceable cutting head (22) is removably attached to the tool holder (24) by the fastening screw (25) located in the head through hole (42) and being threadingly engaged with the holder internal threaded hole (72).

32. The rotary cutting tool (20) according to claim 31, wherein in a locked position of the cutting tool (20), the head axial abutment surface (41) abuts the holder axial abutment surface (71).

33. The rotary cutting tool (20) according to claim 31, wherein in a locked position of the cutting tool (20), each rotationally leading projection flank surface (60L) is spaced apart from a respective

rotationally leading recess flank surface (88L).

34. The rotary cutting tool (20) according to claim 31, wherein in a locked position of the cutting tool (20), the head rearward surface (50) is spaced apart from the female member bottom surface (68).

35. The rotary cutting tool (20) according to claim 31, wherein the rotary cutting tool (20) is a reamer.
